Lab 3: Architectural Pattern Selection — Written Justification

System: Domain & SSL Certificate Expiry Alert System

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Requirement Derivation

Only the system topic was provided for this assignment, with no separate requirements document. Requirements were therefore derived by decomposing the topic name itself: “Domain & SSL Certificate” implies a need to track domain registration and SSL certificate records; “Expiry” implies an ongoing, scheduled monitoring process rather than a one-off check; and “Alert” implies a notification mechanism that warns a user before expiry occurs. These three functional threads (monitoring, scheduling, and alerting), plus the supporting needs for data persistence and a user-facing view, directly produced the five components identified below.

Architecture Selection

We chose a Microservices Architecture for the Domain & SSL Certificate Expiry Alert System.

Reason 1: Independent Scaling of the Monitoring Workload

The Domain & SSL Monitor Service must periodically check WHOIS and SSL expiry data for every registered domain. As the number of monitored domains grows, this checking workload grows with it, while the Web Dashboard UI and Notification Service do not. A microservices architecture allows the Monitor Service to be scaled independently (e.g., by running more instances) without needing to scale the entire application, which a layered or client-server design would require.

Reason 2: Fault Isolation Between Checking and Alerting

Notification delivery (email/SMS) depends on third-party providers that can be slow or temporarily unavailable. Because the Notification Service is a separate component from the Monitor Service, a delay or outage in sending alerts does not block or slow down the scheduled certificate/domain checks. In a tightly coupled layered or client-server design, a failure in one part of the system is more likely to stall the rest.

Security Advantage

Each microservice owns only the credentials it needs: the Notification Service holds the SMTP/SMS API keys, while the Monitor Service and Database hold no notification credentials at all. This isolates sensitive secrets to a single, small service boundary, so a vulnerability in the Web Dashboard UI or the Monitor Service cannot directly expose the notification credentials, reducing the system’s overall attack surface.

Performance Benefit

Because checking and notifying are decoupled services communicating through service APIs, the Scheduler Service can trigger a large number of domain checks across multiple Monitor Service instances instead of processing them one at a time in a single application thread. This keeps the average time-to-detect an expiring certificate low even as the number of monitored domains increases.

Components Identified

- Web Dashboard UI
- Scheduler Service
- Domain & SSL Monitor Service
- Notification Service
- Database

Domain & SSL Certificate Expiry Alert System — Component Diagram (Microservices Architecture)

